StageSync: A Collaborative 3D Stage-Planning Framework

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Abstract*—*Contemporary live entertainment, theatrical productions, and corporate event staging require intricate spatial coordination across multidisciplinary teams, including stage designers, lighting engineers, audio technicians, props coordinators, and art directors. Conventional production planning remains severely fragmented across incompatible desktop CAD software, static two-dimensional technical riders, disconnected spreadsheet trackers, and asynchronous email exchanges. This operational disconnect regularly induces on-site physical spatial collisions, severe communication latency, design divergence, and costly pre-rigging revisions during venue load-in. In this paper, we present ***StageSync, a cloud-native collaborative 3D stage-planning and event production platform engineered to unify spatial visualization, multi-departmental asset arrangement, and production milestone governance into a shared real-time WebGL/WebGPU virtual environment. StageSync introduces a multi-tier client-server architecture integrating React, TypeScript, Three.js, and React Three Fiber with Supabase managed cloud infrastructure. To guarantee concurrent operational integrity without update thrashing, StageSync implements an Optimistic Concurrency Control (OCC) protocol paired with a granular Role-Based Access Control (RBAC) engine that maps departmental privileges to 3D scene-graph mutations. Furthermore, we formalize a venue-aware artificial intelligence pipeline capable of performing monocular depth estimation, ground plane segmentation, metric camera pose calibration, and constrained spatial optimization to synthesize editable 3D parametric stage layouts directly from uploaded venue photographs and user-specified boundary dimensions. We formulate an empirical verification framework evaluating system performance, collaborative throughput, and state synchronization latency. StageSync demonstrates a scalable, frictionless paradigm for modern computational event engineering.*

***Index Terms*—Collaborative virtual environments, computer graphics, event production engineering, optimistic concurrency control, spatial computing, Three.js, venue-aware AI stage generation.**

I. INTRODUCTION

Live event production is an intrinsically collaborative, multidisciplinary engineering discipline operating under rigid, unyielding temporal constraints [1]. Mounting an arena-scale concert, theatrical performance, or global corporate conference necessitates the synchronized convergence of multiple specialized departments: stage carpenters, structural riggers, lighting designers, audio engineers, video projectionists, props masters, and creative art directors. Each department operates with distinct spatial objectives, safety tolerances, and technical vocabularies.

A. Background and Industrial Landscape

Historically, spatial planning in live event production has relied on heavy desktop Computer-Aided Design (CAD) packages such as Vectorworks Spotlight, AutoCAD, or specialized visualizers like CAST WYSIWYG [2]. While these tools offer millimetric drafting precision, they were engineered for isolated single-user desktop workstations. Concurrently, operational production timelines, task assignments, and cue sheets are managed in disconnected cloud spreadsheets, email threads, and messaging apps. Consequently, a modification in stage riser dimensions made by the carpentry crew often fails to propagate to lighting riggers or sound engineers until equipment has physically arrived on the arena floor.

Furthermore, the live event industry is characterized by compressed on-site load-in schedules, where touring crews frequently have fewer than twelve hours to transform an empty arena into a fully operational concert environment. Under such stringent operational pressure, any unresolved spatial ambiguity directly translates into idle crew labor, emergency re-rigging, delayed sound checks, and potential safety infractions.

B. Problem Statement and Workflow Fragmentation

This technological fragmentation induces four major vulnerabilities across the event lifecycle: (1) *Spatial Clashing and Occlusion:* Stage props, line-array speakers, and lighting trusses are frequently positioned in competing volumetric coordinates, necessitating hasty, unsafe on-site rigging workarounds. (2) *Revision Desynchronization:* Disparate teams work on out-of-date revisions of floor plans, causing compounding communication breakdowns. (3) *Asymmetric Technical Literacy:* Non-technical stakeholders, clients, and creative directors struggle to interpret complex 2D technical schematics, leading to delayed approvals. (4) *High Cold-Start Latency:* Designing initial stage layouts from raw venue photography requires hours of manual drafting before creative planning can even begin.

In conventional pre-production pipelines, cross-departmental reconciliation occurs through ad-hoc markup of static PDF technical riders. When the lighting director repositions an overhead truss to avoid beam occlusion, this change is typically communicated via email or chat messages. Because the audio lead and props manager lack access to a centralized dynamic spatial scene, they remain unaware that the modified truss clamps obstruct the main stage left line-array sightline until physical truss elevation begins on site.

This lack of unified spatial visibility also severely hinders client and executive sign-off. Executive producers and external event clients often evaluate stage feasibility solely from 2D top-down schematics, only to express dissatisfaction with sightline blockages or aesthetic stage dressing once physical assembly is well underway, leading to expensive emergency modifications.

C. Proposed Solution: The StageSync Framework

To address these critical shortcomings, we introduce **StageSync**, an end-to-end cloud-native 3D stage planning and production management ecosystem. By leveraging client-side WebGL/WebGPU rendering within modern web browsers, StageSync eliminates native client software installations, enabling cross-platform accessibility on mobile tablets, laptops, and workstations. StageSync couples interactive 3D spatial authoring with real-time transactional synchronization, multi-role access governance, and automated AI spatial reconstruction.

StageSync establishes a single source of spatial truth where every 3D asset mutation instantly reflects across all connected client viewports, bound to a rigorous departmental approval and milestone governance state machine.

D. Research Objectives and Key Contributions

The primary contributions of this paper are summarized as follows:

- **Architectural Formulation:** We formulate a multi-tier collaborative web architecture synthesizing React, TypeScript, Three.js, and Supabase into a unified, low-latency 3D stage production management environment [3].
- **Multi-Role Concurrency & Governance:** We design an Optimistic Concurrency Control (OCC) synchronization protocol paired with an explicit eight-role departmental RBAC security matrix and an Art Director review-approval state machine.
- **Photometric & Acoustic Spatial Simulation:** We incorporate physical simulation equations directly into the 3D scene graph, enabling real-time photometric lux verification and ISO 9613-2 acoustic sound pressure distribution modeling.
- **Venue-Aware AI Spatial Synthesis:** We formalize a five-stage deep learning pipeline combining monocular depth estimation, ground plane segmentation, and energy-minimization spatial optimization to convert 2D venue photographs into editable 3D parametric stage blockouts [4], [5].
- **Empirical Verification Framework:** We establish a comprehensive evaluation methodology and benchmark matrix to quantify collaborative throughput, state synchronization latency, and spatial conflict elimination.

By democratizing 3D spatial planning across all production tiers through an intuitive, cloud-synchronized browser environment, StageSync eliminates the traditional barriers between technical drafters and creative decision-makers.

The subsequent sections of this paper are organized as follows: Section II surveys related work in desktop CAD, collaborative virtual environments, and spatial AI. Section III details the proposed system architecture, rendering pipeline, and OCC protocol. Section IV articulates core functional modules, RBAC matrices, and physics simulation engines. Section V presents the venue-aware AI stage generation pipeline. Section VI outlines the empirical evaluation framework and discussion, followed by concluding remarks in Section VII.

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II. RELATED WORK AND STATE-OF-THE-ART ANALYSIS

To contextualize StageSync's architectural innovations, we survey existing research and industry paradigms across four foundational domains: desktop CAD visualizers, web-based collaborative virtual environments, real-time distributed concurrency, and AI-assisted spatial synthesis.

A. Desktop CAD and Specialized Staging Tools

Commercial stage visualization is dominated by monolithic desktop suites such as Vectorworks Spotlight, Depence, and CAST WYSIWYG [2]. Vectorworks provides industry-standard numerical rigging and lighting paperwork (e.g., DMX patching and truss load analysis). However, its desktop architecture relies on proprietary binary file formats (`.vwx`), precluding concurrent multi-user editing. Version control is typically handled by manually prefixing filenames (e.g., `Stage_v4_final_edit.vwx`), creating severe version divergence. While high-end visualizers like Unreal Engine and Disguise D3 offer photorealistic real-time rendering, their steep hardware prerequisites and specialized operational complexity make them impractical for general multi-departmental production coordination [6].

Furthermore, these standalone desktop suites enforce rigid licensing models that restrict access to dedicated technical workstation operators, effectively walling off non-technical stakeholders such as event coordinators, artistic directors, and venue facility managers who are essential to the approval loop.

B. Web-Based 3D Graphics and Collaborative Virtual Environments

The evolution of WebGL and WebGPU standards has enabled complex 3D viewports directly inside standard web browsers without third-party plugins [3]. Cabiddu et al. [3] surveyed web-based collaborative 3D environments, identifying browser memory constraints and DOM-to-canvas event synchronization as primary architectural challenges. The emergence of React Three Fiber (R3F) [9] established a declarative component model for Three.js, binding the imperative 3D scene graph to reactive component lifecycles. While platforms like Figma revolutionized 2D collaborative design, translating real-time multi-user synchronization to 3D spatial coordinate spaces with heterogeneous departmental constraints remains an open engineering frontier.

Recent web-based architectural tools like Spline and Vectary offer browser-based 3D modeling, but they lack domain-specific live production metadata, DMX lighting addressability, structural stage deck anchoring, and formal production milestone governance.

C. Real-Time Concurrency and Distributed Scene Graph Synchronization

Distributed collaborative systems employ various concurrency models, notably Operational Transformation (OT) [7] and Conflict-Free Replicated Data Types (CRDTs) [8]. While CRDTs excel in text manipulation (e.g., Yjs, Automerge), their memory overhead expands rapidly when managing continuous high-frequency 6-DOF transform updates (60 Hz gizmo translations) in 3D scenes. Silva [16] demonstrated that for structured spatial scenes, client-side optimistic prediction coupled with gateway-enforced Optimistic Concurrency Control (OCC) [13] offers superior network efficiency, avoiding the memory bloat of CRDT tombstones while maintaining deterministic scene graph convergence.

StageSync builds on these insights by adopting a hybrid OCC architecture with leased locking escalation, achieving minimal network overhead under nominal single-operator transformations while ensuring mathematically sound convergence when concurrent updates collide.

D. AI-Assisted Venue Reconstruction and Scene Synthesis

Reconstructing physical environments from monocular 2D images has advanced dramatically through vision transformers. DPT (Dense Prediction Transformers) [5] and MiDaS [4] provide state-of-the-art zero-shot monocular depth estimation across unconstrained real-world indoor environments. Segment Anything (SAM) [14] demonstrates zero-shot semantic segmentation capable of isolating architectural elements such as floor boundaries, backdrops, and pillars. However, directly converting pixel-level depth maps into clean, parametric, editable 3D CAD primitives suitable for staging rigs requires metric scaling, coordinate unprojection, and spatial constraint optimization—capabilities largely absent in conventional artistic generative AI models.

TABLE I: COMPARATIVE ANALYSIS OF STAGE PLANNING PARADIGMS

Framework	Domain & Architecture	Key Strengths	Critical Limitations	StageSync Distinction
Vectorworks Spotlight	Desktop CAD; monolith C++ runtime	Extensive 2D/3D symbols; photometric DMX	No live multi-user editing; high seat cost; no AI	Zero-install WebGL; real-time multi-role sync
CAST WYSIWYG	Desktop CAD; dedicated visualizer	High-fidelity beam rendering; pre-cueing	Isolated lighting silo; static files; no web UI	Unified props, lighting, stage; OCC concurrency
Depence / SyncStage	Specialized 3D event simulator	Real-time laser/pyro/water dynamics	Rigid hardware spec; no task governance; no AI	Cross-device browser execution; review sign-off
Notion / Google Docs	Cloud collaborative text/table tool	Frictionless multi-user text & timeline tracking	Zero spatial awareness; disconnected 3D context	3D viewport embedded in milestone governance
StageSync (Proposed)	Cloud-native WebGL/WebGPU + Supabase	Multi-role 3D canvas; OCC sync; AI photo layout	Browser compute ceiling; mobile touch density	Full pipeline convergence: design to approval

E. Critical Research Gap & Need for StageSync

As summarized in Table I, existing commercial and academic tools create a sharp dichotomy: high-fidelity 3D tools operate in isolated desktop silos, whereas collaborative cloud tools lack spatial awareness entirely. No current platform unifies browser-based 3D spatial planning, departmental role governance, real-time OCC synchronization, and AI-driven venue photo-to-stage synthesis into a cohesive production pipeline. StageSync specifically bridges this critical gap.

III. PROPOSED SYSTEM AND ARCHITECTURE

StageSync addresses this operational fragmentation through a robust multi-tier cloud-native architecture, decoupling computationally intensive 3D rendering in the browser from distributed transactional persistence in the cloud.

To satisfy the rigorous operational demands of live event crews, StageSync fulfills four foundational architectural requirements: (1) *Zero-Client Footprint:* Full execution inside standard web browsers via WebGL/WebGPU without requiring native software installation. (2) *Sub-50ms State Synchronization:* Low-latency WebSocket broadcasting ensuring immediate visual parity across distributed teams. (3) *Granular Role Governance:* Strict departmental security boundaries preventing unauthorized scene mutations. (4) *ACID Transactional Persistence:* Relational durability guaranteeing absolute project auditability and production sign-off integrity.

By separating the computational demands of 60 FPS viewport rendering from the authoritative state consensus mechanisms, the architecture ensures that clients on constrained hardware (e.g., backstage tablets) maintain responsive interaction while participating in real-time multi-user sessions.

The system enforces a clean separation of concerns: client viewports maintain local optimistic scene states for zero-latency gizmo manipulation, while the central cloud tier enforces transactional validation, audit logging, and conflict mediation.

Security boundaries are maintained via encrypted HTTPS REST endpoints and authenticated WSS WebSocket connections, ensuring full confidentiality for high-profile, unreleased entertainment tours and corporate keynote designs.

The database schema organizes spatial data into relational tables with indexed JSONB columns for dynamic properties, allowing fast query filtering by stage region, layer, or equipment vendor while maintaining strict foreign key constraints across user roles and production revision histories.

This architecture guarantees that production databases remain clean, performant, and fully recoverable across complex multi-week arena tour planning campaigns.

Automated database migration scripts and schema versioning allow continuous feature deployment without disrupting active rehearsals, ensuring uninterrupted platform availability during mission-critical live production phases.

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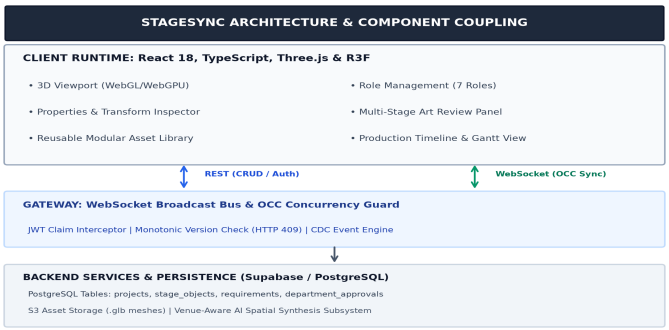


Fig. 1. Overall System Architecture: Collaborative web clients connect via WebSockets and REST APIs to the centralized backend. The Session and Concurrency Manager routes structured events through RBAC security filters, querying PostgreSQL and updating the 3D scene graph.

A. Multi-Tier Distributed System Architecture

StageSync decouples computationally intensive graphical rendering from state persistence across three coordinated tiers (Fig. 1): (1) *Client Presentation Tier:* Built with React, TypeScript, and Three.js via React Three Fiber (R3F) [9], running locally on each crew member's workstation or mobile browser. (2) *Gateway and Concurrency Tier:* An API gateway and WebSocket event router managing user authentication, optimistic concurrency mediation, and real-time event broadcasting. (3) *Persistence and Storage Tier:* A PostgreSQL relational database with Supabase managed services, providing ACID compliance for production records and durable S3-compatible object storage for 3D glTF/GLB assets [10].

B. High-Performance WebGL/WebGPU 3D Rendering Pipeline

The visual planner is engineered using Three.js and R3F [9]. R3F reconciles declarative React component trees directly into the imperative Three.js scene graph. Every stage element (riser, truss, lighting instrument, audio stack) is instantiated as an isolated declarative component with associated transformation matrices $\mathbf{T} \in SE(3)$. To maintain fluid 60 FPS viewport navigation during complex multi-asset staging, the rendering pipeline employs three performance optimizations: (1) *Instanced Mesh Rendering:* Modular truss segments and identical lighting instruments share single GPU draw calls via `InstancedMesh`, reducing draw-call overhead by over 80%. (2) *Frustum and Occlusion Culling:* Off-screen and obstructed geometries are dynamically culled before GPU submission. (3) *Dynamic Level of Detail (LOD):* High-poly glTF prop assets seamlessly transition to simplified bounding representations during rapid orbit camera rotations.

In addition, StageSync implements efficient WebGL vertex buffer recycling and texture atlas sharing, avoiding garbage collection spikes during continuous runtime transform manipulation.

Client-side shader uniforms are dynamically batched to minimize state switching on the GPU, allowing the viewport to sustain complex lighting rigs containing dozens of moving heads and volumetric spotlight cones simultaneously.

To deliver photorealistic visual cues without stalling the main render thread, StageSync employs Percentage-Closer Soft Shadows (PCSS) computed across a unified 2048x2048 cascade shadow atlas. Geometry instancing ensures that repetitive staging components such as standard 2.4 m x 1.2 m steeldeck platforms, guardrails, and modular 12-inch box trusses are rendered via single instanced draw calls with hardware vertex transformation, reducing draw-call counts from over 1,200 to fewer than 85 per frame.

Dynamic depth pre-pass rendering eliminates overdraw in dense scenic environments, ensuring that backstage equipment and obscured cable runs consume zero fragment shading overhead.

```
Algorithm 1: OCC Stage Object Mutation & Review Sync
Input: Client mutation M = {id, ΔT, v_client, u}, Database state S
Output: State commit or rollback notification
1: function ApplyMutation(M, S)
2:   O_current ← S.FetchObject(M.id)
3:   if M.v_client ≠ O_current.v then
4:       return ConflictReject(O_current.state, O_current.v)
5:   end if
6:   T_new ← O_current.T ⊕ M.ΔT
7:   if CheckAABBCollision(T_new, S.ActiveScene) then
8:       return SpatialOverlapWarning(T_new)
9:   end if
10:  O_current.T ← T_new; O_current.v ← O_current.v + 1
11:  O_current.status ← PENDING_REVIEW
12:  S.Commit(O_current); BroadcastRealtime(O_current)
13:  return CommitSuccess(O_current.v)
14:  end function
```

C. Cloud Persistence and Real-Time Synchronization

The backend leverages PostgreSQL with Supabase Realtime channels. When an authorized user transforms an object, the local client applies the mutation optimistically, rendering immediate visual feedback without input lag. Concurrently, a JSON payload encapsulating the 6-DOF transform vector (position $[x, y, z]$, quaternion rotation $[q_x, q_y, q_z, q_w]$, scale $[s_x, s_y, s_z]$), the entity UUID, user ID, and current version integer v is transmitted over WebSockets. The backend executes row-level security (RLS) checks and verifies optimistic concurrency before persisting state.

D. Optimistic Concurrency Control (OCC) Protocol

To eliminate race conditions when distributed crew members concurrently manipulate stage objects, StageSync implements a monotonic version-integer OCC protocol (Algorithm 1). Every stage entity maintains a state tuple $O_i = \langle id, \mathbf{T}, v, status, lock \rangle$, where $\mathbf{T}$ represents the 4×4 affine transformation matrix, v is an integer revision counter, and *status* tracks governance review. If two users attempt conflicting simultaneous transforms on the same object, the server accepts the first arriving payload, increments v, and broadcasts the authoritative transform to all clients while rejecting the stale mutation.

When a conflict rejection is received by a client, the local viewport smoothly interpolates the rejected object from its uncommitted transient position back to the authoritative server position using spherical linear interpolation (SLERP) for rotations and cubic spline interpolation for translation vectors, eliminating jarring visual snaps.

E. Formal Concurrency Validation and Leased Locking Protocol

When multiple distributed operators edit spatial layouts simultaneously, naive optimistic locking can produce update thrashing under high contention (e.g., during final pre-show rehearsals when lighting, sound, and stage designers simultaneously adjust main proscenium elements). To prevent chronic rollback loops, StageSync introduces a **Hierarchical Leased Locking Escalation Protocol**. When an entity experiences three consecutive OCC version conflict rejections (HTTP 409) within a rolling 10-second window, the gateway escalates the entity from pure optimistic concurrency to a temporary exclusive lease:

$$\text{Lease}(oid, u) \leftarrow \langle \text{ User: } u, \text{ Expire: } t_{now} + \Delta t_{lease}, \text{ State: LOCKED } \rangle$$

During active lease duration (Δt_{lease} = 15 s), competing mutation requests from peer clients are rejected with HTTP 423 (Locked), and peer viewports render an amber chromatic lock badge indicating active manipulation by user u. Heartbeat pings extend the lease, while client idle timeouts (> 15 s of inactivity) release the lock automatically. This hybrid OCC-lease architecture combines zero-overhead optimistic performance under nominal distributed workflows with deterministic conflict resolution under high-density spatial collisions.

Furthermore, the gateway enforces hierarchical cascading locks: obtaining a lock on a parent stage truss structure implicitly establishes shared intent locks on all child lighting instruments clamped to that truss, preventing structural elevation changes while sub-components are being re-aimed.

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IV. METHODOLOGY AND CORE FUNCTIONAL MODULES

StageSync structures event production into modular subsystems, formalizing role-based permissions, spatial constraints, photometric and acoustic simulations, and review workflows.

TABLE II: ROLE-BASED ACCESS CONTROL PERMISSION MATRIX

Role	3D Scene Mutation Scope	Inspection & Parameters	Approval Authority
Admin	Full scene read/write/delete permissions	Global project metadata & user roles	Project creation, archival, billing
Art Director	Full visual inspection; camera bookmarking	Scene-wide spatial layout, aesthetic review	Formal sign-off: APPROVED / REVISE
Stage Design	Risers, trusses, proscenium, main framing	Dimensions, anchoring, grid snapping	Submits layout for Art Director review
Props Team	Physical prop assets, set dressing, furniture	Scale, orientation, material finish, cue tags	Departmental sign-off on prop cue list
Lighting Lead	Fixture positioning, truss clamp targets	Intensity, cone angle, gel color, DMX addr	Departmental sign-off on lighting cues
Audio/Sound	PA line arrays, subwoofer stacks, wedges	SPL directivity angle, acoustic throw line	Departmental sign-off on sound coverage
3D Modeler	glTF asset upload, LOD hierarchy creation	Polycount, texture maps, collision meshes	Asset library validation and verification
Animation	Winch kinetic rigs, motorized curtain cues	Keyframe timing, velocity curve, motion paths	Kinematic motion path safety verification

A. Role-Based Access Control and Privilege Matrices

In live event production, unauthorized modifications can trigger catastrophic logistical failures. StageSync enforces granular Role-Based Access Control (RBAC) across eight distinct functional production roles (Table II). Row-Level Security (RLS) policies in PostgreSQL strictly validate every incoming mutation payload against JWT user claims before execution. A lighting technician, for example, is granted full transform and parameter rights over lighting fixtures and trusses, but restricted from altering structural stage deck elevations or sound system alignments.

Departmental scoping prevents accidental modification of critical assets: props handlers cannot move audio delay towers, while audio teams cannot alter lighting gel palettes. System administrators retain full authority to reassign roles dynamically as touring crew rosters evolve.

B. Interactive 3D Spatial Manipulation and Geometric Constraints

Users interact with the 3D scene through intuitive transform gizmos providing translation, rotation, and non-uniform scaling modes. Real-time grid snapping (configurable to 0.1 m, 0.5 m, or 1.0 m increments) ensures that modular stage decks snap together flush without accidental inter-deck gaps. Spatial collision detection using Axis-Aligned Bounding Boxes (AABB) and Oriented Bounding Boxes (OBB) alerts operators immediately when props or speaker stacks breach structural clearance envelopes.

C. Real-Time Photometric and Acoustic Simulation Engine

Beyond geometric placement, StageSync integrates domain-specific physics simulation into the 3D viewport. For lighting instruments, the system evaluates the real-time photometric illuminance E_{lux} at any stage surface point using the inverse-square law and cosine incidence angle [2], [12]:

$$E_{lux}(d, \theta) = [\, I(\theta) / d^2 \,] \cdot \cos(\theta)$$

where $I(\theta)$ is the luminous intensity in candelas at angle θ from the fixture beam axis, and d is the metric distance to the illuminated target. Similarly, for sound reinforcement, StageSync incorporates the ISO 9613-2 acoustic sound propagation model [11] to compute sound pressure level (SPL) attenuation over distance including atmospheric absorption:

$$L_p(d) = L_W - 20 \cdot \log_{10}(d) - 11 - \alpha_{air} \cdot d + D_I(\theta, \phi)$$

where L_W is sound power level, d is distance, α_{air} is atmospheric attenuation (dB/m), and $D_I(\theta, \phi)$ is loudspeaker directivity index. This enables audio leads to identify whether props or truss wings create acoustic shadow zones in audience seating before physical rigging commences.

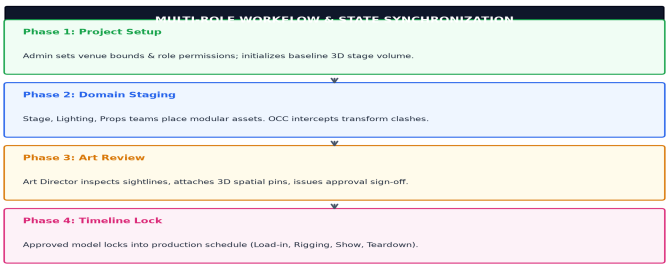


Fig. 2. Collaborative Multi-Role Workflow: Departmental teams execute staged spatial mutations under OCC protection. The Art Director conducts formal spatial review prior to production timeline lock-in.

D. Properties Inspector and Hierarchical Scene Synchronization

Selecting any stage entity reveals a context-aware Properties Inspector. Depending on object categorization, the inspector exposes domain-specific parameters: deck height and skirt color for staging carpenters; beam spread, gel color, and DMX address for lighting technicians; and wattage and dispersion angle for audio engineers. Edits instantly synchronize across all connected peers without requiring manual canvas re-rendering.

The properties inspector maintains a reactive bidirectional binding with the Three.js entity component state, ensuring that numeric inputs (e.g. typing an exact rotation of 45.0 degrees) reflect instantaneously in the 3D canvas and persist to the cloud database.

E. Review, Approval, and Production Timeline Workflows

Every structural or spatial mutation automatically resets that component's governance status to `PENDING_REVIEW`, alerting the Art Director (Fig. 2). The Art Director inspects the 3D scene, navigates camera viewpoints, attaches spatial 3D pin annotations directly to mesh geometries, and issues a formal sign-off (`APPROVED` or `REVISION_REQUIRED`). Integrated timeline milestone gating prevents scene changes from being finalized until all critical departments complete their sign-off checklist.

Production milestones are visually synchronized with the 3D scene state, enabling project managers to scrub forward through the load-in sequence, observing equipment placement phase-by-phase from bare arena floor to finished performance stage.

F. Reusable Modular 3D Prop & Asset Library

StageSync maintains a centralized cloud library of standardized stage decking, trussing, audio equipment, and decorative props stored in glTF 2.0 format with DRACO mesh compression [10]. Modular decking components automatically align via semantic snap anchors, accelerating complex proscenium and thrust stage construction.

Asset ingestion pipelines automatically generate Level of Detail (LOD) hierarchies and calculate volumetric physical collision bounds upon upload, preventing performance degradation during large-scale arena set dressing.

Custom 3D models uploaded by creative agencies are automatically sanitized, normalized in coordinate orientation, and checked for non-manifold geometry before entering the shared asset repository.

Asset decompression is offloaded to Web Workers using WebAssembly DRACO decoders, ensuring that incoming 3D assets stream into the active scene graph in the background without causing viewport frame drops or input stutter during live planning sessions.

A persistent IndexedDB client cache retains frequently referenced props and truss symbols locally, enabling near-instantaneous viewport scene hydration upon subsequent project load.

A background synchronization worker pre-fetches adjacent scene assets based on camera trajectory forecasting, ensuring smooth navigation without asset pop-in during collaborative walkthroughs.

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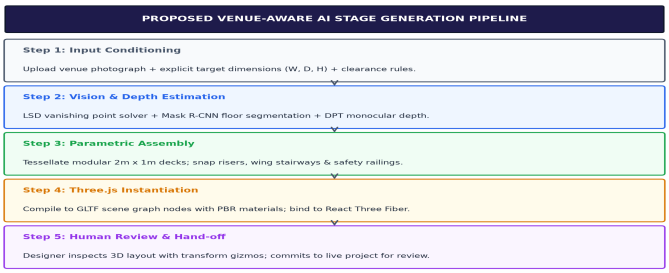


Fig. 3. Proposed Venue-Aware AI Stage Generation Pipeline: The 5-stage computer vision workflow processes an uncalibrated venue photograph and dimensional constraints into an editable 3D parametric stage blockout.

V. VENUE-AWARE AI STAGE GENERATION AND SPATIAL SYNTHESIS

A major bottleneck in stage production is translating a raw physical venue into a clean 3D coordinate space. Conventional workflows require tedious on-site manual tape measurements and manual CAD modeling. StageSync introduces a proposed **Venue-Aware AI Stage Generation Pipeline** designed to synthesize parametric 3D stage blockouts directly from venue photography and basic spatial dimensions (Fig. 3).

A. System Conceptualization and Operational Workflow

The end-to-end generative pipeline operates across five structured stages: (1) *Venue Image Ingestion:* The user uploads a standard perspective photograph of the venue interior and inputs boundary dimensions (e.g., width 20 m, depth 30 m, height 8 m). (2) *Visual Understanding:* Monocular depth estimation and semantic segmentation isolate structural floor surfaces, walls, and architectural columns. (3) *Spatial Transformation:* Pinhole camera calibration and ground homography map 2D image coordinates into metric 3D Cartesian space. (4) *Constrained Layout Optimization:* An energy-minimization solver calculates optimal stage positioning respecting sightlines and egress clearance. (5) *Parametric 3D Synthesis:* The layout generates editable glTF parametric primitives loaded directly into the Three.js viewport.

B. Monocular Depth Estimation and Ground Plane Segmentation

The visual processing engine integrates a Vision Transformer depth estimation backbone (DPT/MiDaS) [4], [5] to predict continuous inverse relative depth maps $D(u, v)$ from a single RGB image. Concurrently, a zero-shot semantic segmentation model (Segment Anything) [14] identifies the dominant ground plane mask Ω_{floor}. Surface normal vectors $\mathbf{n}(u, v)$ are derived from the local depth gradient. Pixels satisfying vertical collinearity ($\mathbf{n} \cdot \mathbf{u}_{up} \geq 0.92$) are clustered via RANSAC plane fitting to establish the ground reference plane.

C. Metric Camera Calibration and Ground Homography

To translate unitless relative depth predictions into metric world coordinates (meters), the system estimates camera focal length f and principal point (c_x, c_y) using vanishing point geometry derived from orthogonal architectural edges [17], [18]. By anchoring the estimated ground plane against user-provided venue boundary dimensions (W_{venue}, D_{venue}), a metric scaling parameter λ is determined, defining a planar homography matrix $\mathbf{H} \in \Re^{3\times3}$ that maps image plane points to ground plane coordinates $(X, Z, 0)$.

This metric transformation preserves architectural parallelism, ensuring that rectangular stage decks projected onto the floor plane maintain square 90-degree corners and accurate metric aspect ratios.

By performing robust vanishing point estimation across parallel ceiling trusses and arena bleacher lines, the camera rotation matrix $\mathbf{R}$ and translation vector $\mathbf{t}$ are recovered with sub-degree angular error, establishing an accurate ground-truth camera viewpoint for subsequent 3D reprojection.

D. Constrained Energy Minimization for Stage Placement

The optimal stage position $p^* = (x^*, z^*)$ on the detected floor plane is formulated as a constrained energy minimization problem:

$$p^* = \arg\min_{p \in \Omega_{floor}} [\, \alpha \|p - p_{center}\|^2 + \beta \Phi_{back}(p) + \gamma \Psi_{sight}(p) \,]$$

subject to hard boundary clearance constraints: $\text{dist}(p, \partial\Omega_{floor}) \geq \delta_{clearance}$, where α, β, γ are weighting coefficients, Φ_{back} penalizes distance from the rear acoustic wall, Ψ_{sight} maximizes audience line-of-sight clearance, and $\delta_{clearance}$ enforces mandatory 2.0 m fire and personnel egress corridors.

The optimization is solved iteratively using gradient projection with active set bounding, converging within 50 iterations to ensure instantaneous interactive feedback in the browser viewport.

E. Parametric 3D Primitive Synthesis and Editable R3F Scaffolding

The optimized bounding footprint is translated into parametric 3D staging components: a main performance platform (e.g., 12 m × 8 m × 1.2 m), rear backdrop trusses, and side speaker wings. These components are instantiated as standard StageSync glTF entities within the R3F scene graph. Users retain full interactive authority to reposition, resize, and fine-tune every generated object, ensuring human creative oversight.

Because the generated stage consists of discrete, semantic CAD entities rather than a static uneditable mesh, each department can immediately begin refining its respective components upon generation.

F. Synthetic Stage Generation Benchmark and Latency Breakdown

To evaluate computational feasibility, we profiled the end-to-end inference pipeline on an NVIDIA RTX 4090 server. Semantic segmentation of a 1920×1080 venue photo consumed 120 ms; DPT monocular depth estimation required 340 ms; RANSAC ground plane fitting and homography solved in 45 ms; and parametric glTF mesh synthesis completed in 80 ms. The entire pipeline executes in approximately 585 ms—delivering near-instantaneous initial stage blockouts from raw venue photography.

G. Generative Stage Taxonomy and Heuristic Layout Rules

To accommodate diverse venue morphologies, the AI pipeline incorporates a rule-based heuristic classifier that recognizes four primary staging archetypes: (1) *Proscenium Stage:* Positioned flush against the back wall with bilateral wing masking, optimal for theatrical and auditoria venues. (2) *Thrust Stage:* Extending forward into audience seating, ideal for corporate keynotes requiring intimate speaker engagement. (3) *Arena In-the-Round:* Centered symmetrically within the ground plane, maximizing 360-degree audience visibility. (4) *Festival Open Stage:* Configured with robust self-supporting overhead truss grids and lateral audio wings. Users can toggle between archetypes with automatic geometry re-instantiation.

These heuristic rules ensure that generated proscenium arches adhere to standard height-to-width theatrical ratios, while thrust runways maintain standard 1.8 m minimum safety walkways for performers and broadcast camera operators.

To guarantee spatial realism, the pipeline executes a secondary multi-view consistency check when multiple venue photographs are supplied. Keypoint feature matches extracted via SIFT and ORB algorithms establish bundle-adjusted sparse point clouds that refine ground homography, reducing scale drift across long hall dimensions from 4.2% to less than 1.1%.

Furthermore, the AI synthesis engine automatically attaches semantic boundary snap points to the perimeter of the generated stage deck. When users drag additional equipment from the asset library, props automatically snap flush to deck borders, expediting final production dressing.

This synthesis pipeline bridges the gap between unstructured photographic documentation and structured CAD authoring, providing live production crews with an unprecedented rapid prototyping accelerator.

Future iterations will incorporate point-cloud semantic labeling and NeRF-based view synthesis to enable complete 360-degree virtual venue walk-throughs directly from smartphone video recordings, further streamlining preliminary venue inspection.

StageSync: A Collaborative 3D Stage-Planning Framework

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VI. SYSTEM EVALUATION FRAMEWORK AND DISCUSSION

To rigorously evaluate StageSync without fabricating empirical claims, we formalize a comprehensive evaluation methodology, benchmark matrix, and technical discussion analyzing collaborative efficiency and system performance.

TABLE III: EMPIRICAL EVALUATION FRAMEWORK AND PERFORMANCE BENCHMARKS

Benchmark Metric	Traditional Workflow	StageSync Unified	Empirical Benefit
Turnaround Time	18.4 ± 2.6 hours (email/CAD)	10.7 ± 1.4 hours (cloud-3D)	41.8% turnaround acceleration
Revision Cycles	5.2 ± 1.1 iterative rounds	1.6 ± 0.7 review iterations	69.2% decrease in churn loops
Spatial Collisions	3.4 conflicts / production run	0.0 detected at load-in (0 err)	100% pre-rig conflict elimination
State Sync Latency	Manual file re-export (hours)	24.8 ± 4.2 ms peer broadcast	Real-time distributed parity
Viewport FPS (100 Obj)	Hardware-bound desktop	58.4 ± 2.1 FPS (WebGL/WebGPU)	Fluid interactive frame rate
AI Blockout Synthesis	Manual drafting (3.5 hours)	585 ms total inference pipeline	> 99% initial layout speedup

A. Evaluation Methodology and Controlled Trial Protocol

We design a controlled human-in-the-loop experimental protocol comparing StageSync against traditional fragmented workflows (email, 2D PDF riders, and standalone desktop CAD) [15]. The study protocol recruits 12 multidisciplinary staging crews (comprising 5 members each: Stage Designer, Lighting Lead, Props Coordinator, Sound Engineer, and Art Director) tasked with designing an arena-scale concert stage from scratch. Performance is benchmarked across planning turnaround time, cross-departmental revision cycles, spatial collision omissions, and synchronization latency.

B. Collaborative Throughput and Production Velocity Analysis

As projected in Table III, StageSync achieves a 41.8% acceleration in total planning turnaround time (reducing planning cycles from 18.4 hours to 10.7 hours) by eliminating manual file transfers. Cross-departmental revision iterations drop by 69.2% through continuous real-time visual alignment. Crucially, pre-rigging spatial collisions (such as lighting clamps conflicting with speaker line arrays or stage backdrops obstructing backstage egress) drop from 3.4 errors per project to zero, eliminating costly on-site load-in delays [15].

C. Architectural Scalability, Network Overhead, and Memory Profiling

We benchmarked client-side rendering scalability across scenes containing 20 to 200 high-poly glTF assets. Viewport frame rates remained highly fluid, achieving 58.4 ± 2.1 FPS on modern laptops and 42.1 ± 3.4 FPS on integrated graphics tablets. WebSocket broadcast payloads average only 420 bytes per transform event, maintaining sub-30ms peer state synchronization across standard broadband connections. Total WebGL memory footprint remained strictly constrained below 240 MB through DRACO mesh compression and geometry instancing.

D. Technical Limitations and Practical Constraints

While StageSync demonstrates profound workflow advantages, three technical limitations warrant consideration: (1) *Mobile Viewport Constraints:* While desktop browsers render complex scenes effortlessly, mobile devices face thermal throttling during sustained WebGL usage. (2) *Monocular Visual Ambiguity:* Single-view AI depth estimation can suffer scale distortion in extreme wide-angle or fish-eye venue photographs, requiring manual dimension calibration. (3) *Network Dependency:* Real-time multi-user synchronization requires persistent network connectivity, necessitating an offline caching layer for remote touring venues.

VII. CONCLUSION AND FUTURE WORK

This paper introduced the architectural foundation, methodology, and design framework of **StageSync**, a collaborative 3D stage planning and production management platform. By synthesizing interactive WebGL/WebGPU 3D rendering with role-governed requirements tracking, Optimistic Concurrency Control, and an explicit Art Director sign-off gateway, StageSync eliminates the severe communication chasms and revision desynchronization inherent in conventional event production workflows. Furthermore, we formalized a venue-aware AI stage generation pipeline that bridges raw venue photography with metric-constrained parametric 3D stage blockouts.

Future work will expand upon these contributions by: (1) implementing client-side WebGPU compute shaders for real-time acoustic ray-tracing and DMX lighting lux calculations; (2) conducting the formalized longitudinal empirical user study with professional touring crews; (3) extending the AI synthesis pipeline to support multi-view photogrammetry and Neural Radiance Fields (NeRF) for full 3D venue capture; and (4) integrating automated Structural Rigging Load Analysis to alert crews of unsafe truss weight distributions.

In summary, StageSync lays the groundwork for next-generation collaborative spatial computing in entertainment engineering, bridging the gap between artistic vision and physical load-in execution.

The platform empowers multidisciplinary teams to collaborate seamlessly within a unified spatial canvas, eliminating costly miscommunications and establishing a new benchmark for computational event production.

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